🧬 PyMutScan — Advanced DNA Mutation Detector

Sequence alignment · Variant calling · Codon translation · Primer design · Drug resistance · Mutation heatmap · Patient records

› 👤 Virtual Patient Manager — Create or Load a Patient Profile

📄 Reference — BRCA1

```
ATGGATTTATCTGCTCTTCGCGTTGAAGAAGTACAAAATGTCATTAATGCTATGCAGAA
```

67 bp demo segment · Source: NCBI GenBank `NM_007294.4`

⌄ 📖 View Full Gene Info & Sources

Full Gene Length

81,189 bp

Chromosomal Location

Chr 17q21.31

Function: DNA damage repair, tumor suppression

RefSeq Accession: `NM_007294.4`

NCBI Gene ID: `672`

⚠️ **Note:** PyMutScan uses a 67 bp demo segment for educational purposes. Full clinical sequences available at NCBI.

🌐 NCBI Gene 📄 FASTA 🔬 ClinVar 📚 OMIM

🔬 Query Sequence (Patient)

⚡ Load Sample with Mutations

Paste DNA sequence or upload FASTA above (A, T, C, G only)

ATGGATTTATCAGCTCTTCGCGTTGATGAAGTACAAAATGTCATGAATGCTATGCA
GAAAATCTTAG

67 bp entered

🔍 Run Mutation Analysis

✅ Analysis complete — 3 variant(s) detected

Total Variants	Sequence Identity	🔴 Pathogenic	🟡 Uncertain	🟢 Benign
3	95.5%	1	1	1

🔬 Alignment 📋 Mutations Table 🧪 Sequence Stats & GC 🔤 Codon Translation 🗺️ Mutation Heatmap 🧲 Primer Design ✂️ Restriction Enzymes 💊 Drug Resist

Sequence Statistics

Reference Sequence

GC Content

34.33%

AT Content

65.67%

Query (Patient) Sequence

GC Content

35.82%

AT Content

64.18%

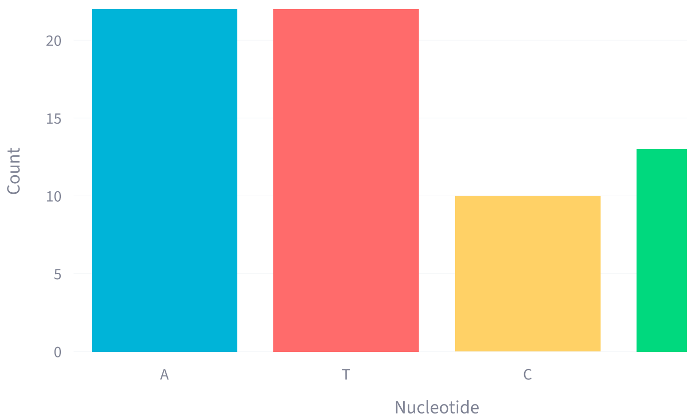
Length

67 bp

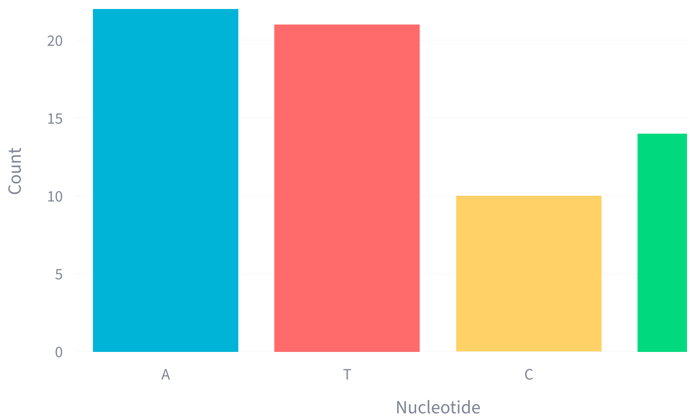
Length

67 bp

Nucleotide Distribution — Reference

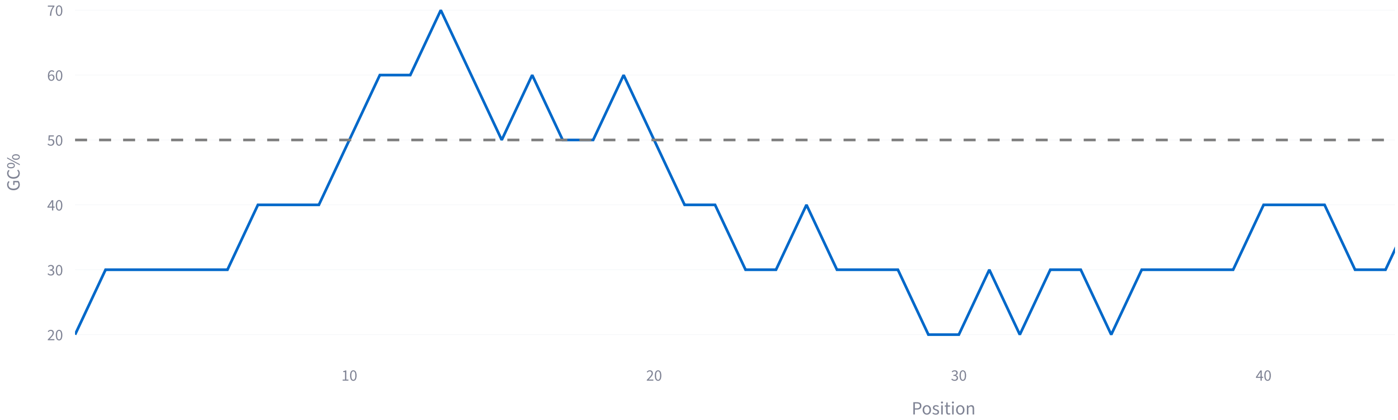


Nucleotide Distribution — Query



GC Content Sliding Window (10 bp)

GC% Along Query Sequence



Patient Records & Database

Persistent SQLite storage — all patient profiles and analysis results are saved locally in `pymutscane_patients.db`

Patient Profiles | Analysis History | Export Data

No patients saved yet. Use the **Virtual Patient Manager** above to create one.